Capability Emergence Can Be Forecast:
Per-Seed, In Advance, With Calibrated Intervals, Certified False Alarms, and a Blind Pre-Registered Gate

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Abstract

Emergent capabilities are widely treated as unpredictable: aggregate loss improves smoothly while specific abilities appear abruptly. Prior work offers early-warning *indicators* but never scores them as *forecasts* — no lead time at a controlled false-alarm rate, no calibration, no negative controls, no blind validation. We supply that discipline and show that, in controlled settings spanning grokking model systems and small language models, emergence *timing* is forecastable per training run, in advance, with calibrated uncertainty. On language models: across 30 transformers at identical configuration — where the strongest published baseline predicts a single constant [1] — the formation time of the previous-token head (the mechanistic precursor of induction [2]) forecasts each seed's induction emergence at Spearman $\rho = 0.977$ (95% CI [0.911, 0.995]) with median lead 975 steps (∼15% of training). A best-case training-loss rule *ties* that ranking ($\rho = 0.977$) but with 50-step median lead — a nowcast, not a forecast. Split-conformal intervals of width 300 steps, anchored at the precursor alarm, covered 15/15 held-out seeds at 90% nominal. The entire rule was then frozen verbatim — alarm threshold, offset 975, quantile 150, commit-stamped — and subjected to a blind pre-registered gate on a never-seen configuration (wider model, sparser repetition): it covered 10/10 runs (bar 7/10), alarmed pre-event 10/10, and false-alarmed on 0/3 fresh negatives; the precursor-to-emergence gap was approximately configuration-invariant (975→1,012 steps). All false-alarm claims are certified against 15+3 *manufactured capability-blocked negatives* (data-ablated and architectural) — a resource we argue is mandatory for capability monitoring and that no public suite provides. Public-checkpoint evidence (Pythia): the precursor leads the induction cliff by a full checkpoint stage in all 5 suites probed, and all five cliff at the same token count, confirming that public suites cannot test per-seed forecasting. The grokking half of the paper (466 runs) contributes the evaluation methodology and its hard lessons: a robustness-versus-false-alarm tradeoff (the context features that suppress false alarms are exactly what breaks under intervention), mechanism-factored gates that resolve it, non-transfer of fitted calibration across regimes, and a label-free probe family that died under pre-registered confirmation. Three kill criteria fired and are reported. Every prediction, freeze, and verdict in both halves is commit-stamped before its data existed. Scope: named capabilities with known mechanistic precursors, small models; we state what stands between this and frontier deployment.

1 Introduction

When a capability emerges during training, the training curve rarely announces it in advance: loss declines smoothly while the ability arrives abruptly [2, 3]. For frontier-scale systems this unpredictability is a safety problem with a name — labs want to know what a model will be able to do *before* it can do it. A literature of early-warning signals has grown in response: loss-curve oscillations that predict whether grokking will occur [4], spectral observables that show precursors of phase transitions [5], geometric stage detection [6], and scale-axis extrapolations [7, 8]. What none of this work does is score a *forecast*: state, in advance, when a capability will arrive; attach calibrated uncertainty; measure lead time at a controlled false-alarm rate against negatives where the capability never arrives; and validate the frozen system blind. Indicators without that discipline are weather lore, not weather forecasting.

This paper does two things. **First**, it builds the missing evaluation discipline on a grokking model system — 466 training runs across two domains with interventions that shift emergence timing $\sim$100$\times$ — and reports what the discipline kills: single-signal alarms die on look-alike negatives; the context features that control false alarms are exactly the features that break under interventional shift (a robustness–false-alarm tradeoff); fitted calibration does not transfer across regimes; and an initially spectacular label-free probe family failed its pre-registered confirmation. **Second**, it carries the surviving methodology to language models and obtains the result the discipline was built for (Figure 1): per-seed forecasts of induction-head emergence from a mechanistic precursor, with calibrated intervals, certified false alarms, and a blind pre-registered transfer gate passed at ceiling.

Contributions, each with its adversarial control:

1. **A forecasting benchmark and protocol for emergence timing**: event definitions robust to outliers, lead time at capped false-alarm rate, champion selection on training data only, (correlation, lead) reported as a pair — plus a *negative taxonomy* (budget-censored vs. structurally blocked vs. *manufactured*) and a minimum-negative-count rule learned the hard way.

2. **The per-seed result** (Sections 5.2–5.3): at fixed configuration, where the config-equation baseline of Aoyama and Wilcox [1] is a constant by construction, precursor formation time forecasts emergence at $\rho = 0.977$ with 975-step median lead; a best-case loss rule ties the correlation with 50 steps of lead. Warning time lives in *where you anchor*, and only the circuit precursor anchors early.

3. **Manufactured negatives** (Section 5.2): capability-blocked training runs — data-ablated (repetition stripped) and architectural (one layer) — enabling the first certified false-alarm rates for emergence forecasting (0/10 and 0/10 on data-ablated, 0/5 on architectural, 0/3 blind).

4. **The blind gate** (Section 5.4): the full rule frozen verbatim and commit-stamped, then tested on a never-seen configuration: 10/10 interval coverage, 10/10 pre-event alarms, $\rho = 0.988$. The precursor-to-cliff gap is approximately config-invariant across the shift we tested — an unexplained regularity we flag as an open question.

5. **The kill ledger** (Section 6): three pre-registered kill criteria fired across the program and are reported with their mechanisms, alongside every failed prediction. The positive results above survived the same regime that produced these deaths; that, not any single number, is the paper's claim to trust.

Everything is pre-registered in a public commit chain (every freeze precedes its data; Section 9), and the corpus, fleets, and harness are released as a benchmark.

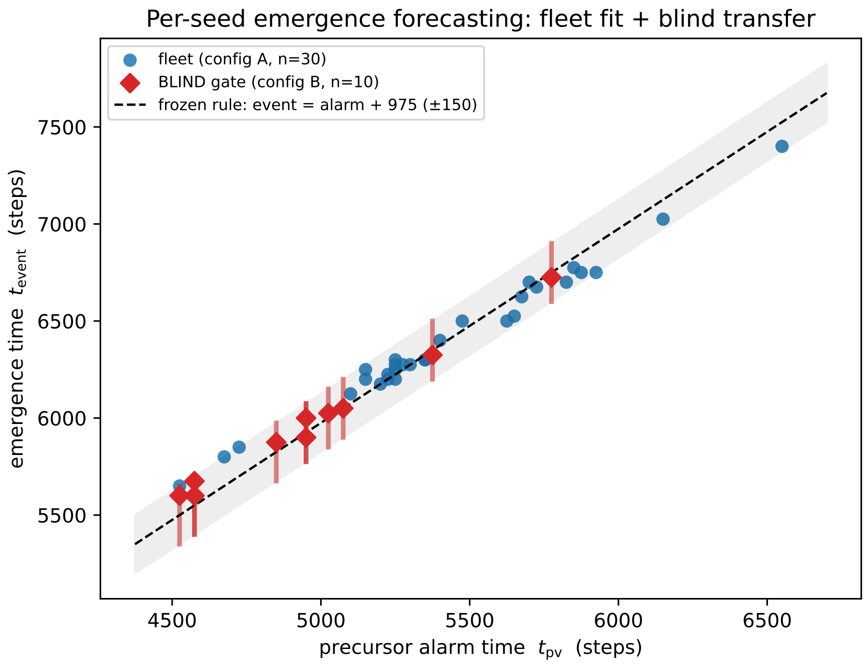

Figure 1: **The central result.** Precursor alarm time vs. emergence time. Blue: the calibration fleet (30 seeds, config A) — Spearman $\rho = 0.977$. The dashed line and band are the *frozen* rule (event = alarm $+975 \pm 150$). Red diamonds with bars: the blind gate — 10 fresh seeds on a never-seen configuration (wider model, sparser repetition), scored once against the frozen rule: 10/10 covered. Negatives (18 manufactured capability-blocked runs, not shown) produced zero false alarms.

2 Related work

Notsawo et al. [4] predict *whether* grokking will occur from early loss-curve oscillations — a scalar whether-classifier; its oscillation feature is one of our baselines. Anonymous et al. [5] derive spectral early-warning indicators (2-RDM heat capacity) across four transition settings but report no lead time, calibration, false-alarm rate, or baseline race; our label-free rung (Section 6) shows concretely why unscored spectral indicators can mislead: ours died under confirmation. Hoogland et al. [6] detect developmental stages retrospectively via the local learning coefficient. Aoyama and Wilcox [1] predict induction-head formation time from pre-training config (batch, context, bigram statistics) — powerful, but constant per config; our fleet result answers precisely the variance their equation cannot (per-seed timing at fixed config), and our Pythia analysis confirms their account across public suites (all five cliff at the same token count). Scale-axis emergence prediction [7, 8] forecasts across model size, orthogonal to our within-run time axis. Olsson et al. [2] established the induction-head phase change and its previous-token precursor; Barak et al. [9] and progress measures [10] established hidden progress; we turn these observations into scored forecasts. The grokking substrate builds on [11–13] and our companion papers [14–16], which established the representational-timing mechanism and the norm clock that our two-gate forecaster exploits.

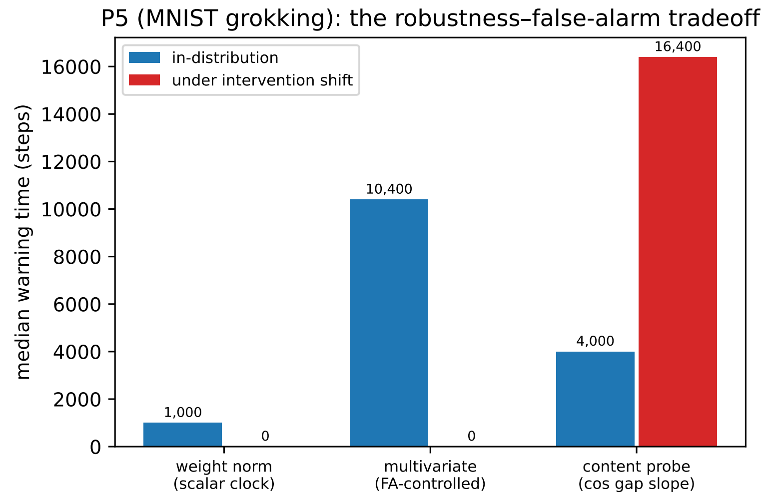

Figure 2: **The robustness–false-alarm tradeoff** (MNIST grokking; median warning time at $\leq 5\%$ FA, higher is better). The scalar clock (weight norm) forecasts in-distribution but collapses when interventions decouple the norm from the outcome. The multivariate forecaster that best controls false alarms in-distribution (10,400 steps, 60% of the delay) collapses hardest under shift — its norm-context features are exactly what the intervention breaks. The bare content probe survives the shift (16,400 steps) but is FA-fragile in corpora with look-alike negatives.

3 Forecasting emergence: definitions and discipline

A **run** logs behavioral and internal signals at fixed cadence. An **event** is the first crossing of an absolute behavioral criterion robust to outliers (never a fraction of a run's own maximum — a rule adopted after that operationalization failed twice; Section 6). A **forecaster** observes the log prefix and may **alarm** once; its **lead** is $t_{\text{event}} - t_{\text{alarm}}$ (alarms after the event score zero; no alarm on an eventing run is a miss). **False-alarm (FA) rate** is the fraction of *negative* runs — runs where the capability never arrives — that alarm. Thresholds and champions are selected on training runs only, under an FA cap ($\leq 5\%$), with at least five negatives on every side of every split (a minimum learned from a degenerate fit, Section 6). Negatives come in three kinds: *budget-censored* (would event with more budget — these legitimately attract alarms), *structurally blocked* (wrong learned structure), and *manufactured* (constructed so the capability cannot form). Forecast quality is always a *pair* — (ranking skill, lead) — because a rule can rank perfectly by detecting the event as it happens; and point forecasts carry split-conformal intervals whose empirical coverage is reported. Blind validation means: every constant frozen and commit-stamped, then fresh runs, one scoring pass.

4 Part I: the grokking benchmark, and what the discipline kills

We built the protocol on 466 pre-existing grokking runs (modular arithmetic transformers; MNIST MLPs in the Omnigrok regime) whose companion-paper interventions (contrastive priors, norm clamps) shift emergence timing up to $\sim 100\times$ and — crucially — *decouple* the dominant scalar clock (weight norm) from the outcome. Full protocol and verdicts are in the repository (`plan_p5.md`, `P5_RESULTS.md`); four results shape everything downstream.

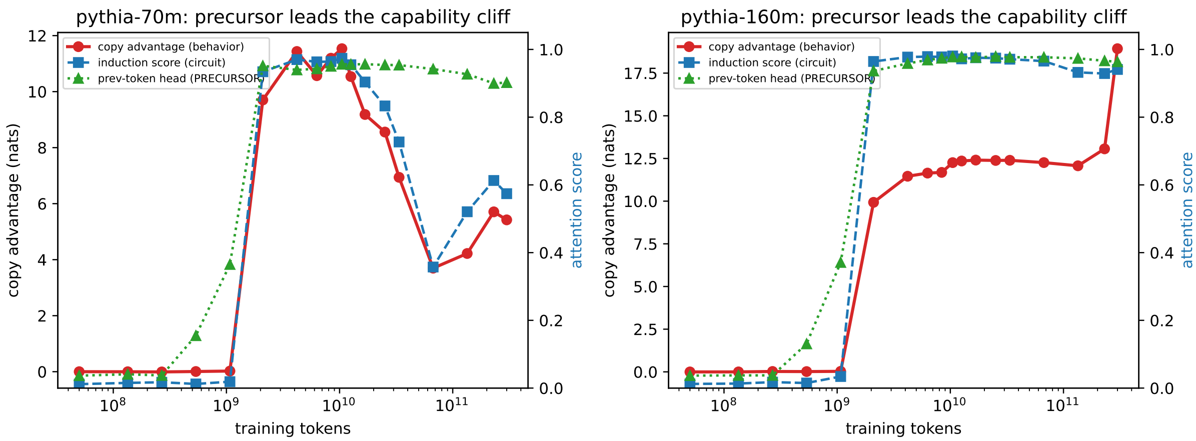

Figure 3: **Public Pythia checkpoints (models we did not train).** The previous-token head (green) rises a full public-checkpoint stage before the induction-head score (blue) and the behavioral copy advantage (red) cliff together. Left: pythia-70m; right: pythia-160m. The 70m model also partially *loses* induction late in training (prefix 0.93→0.36 at 67B tokens) — capability regression exists and is monitorable.

(i) In-regime forecasting works. A multivariate forecaster with mechanism-motivated features achieved 10,400-step median lead (60% of the delay) on held-out seeds at zero structural false alarms, and 18,000 steps prospectively on pre-registered fresh runs including a never-seen norm pin (bar 5,200) — all four prospective predictions passed.

(ii) The robustness–false-alarm tradeoff (Figure 2). Trained on control runs only and tested on intervention arms, the scalar norm clock collapses (median lead 0; the interventions hold the norm where controls never generalize), and the FA-controlled multivariate collapses with it — *because* the context features that suppress false alarms are the features the intervention breaks. The bare content probe survives (4,400–16,400 steps). A monitor tuned quiet on its training distribution goes blind exactly when the recipe changes.

(iii) Mechanism-factored gates resolve the tradeoff where mechanism is known. Composing a content gate with an *a-priori* norm-viability window (constants from the companion papers, not fit) retains 12,400 steps (53% of the delay) under the same shift at zero miss. The portable ingredient is mechanism, not fitted calibration: fitted thresholds transferred across domains in neither direction (kill K5).

(iv) The discipline kills seductive numbers. An algorithmic-domain corpus with structure-complete-but-blocked negatives admitted *no* forecaster at 5% FA (kill K1/K8): trap negatives carry more "progress" signal than true positives show before their events. And a label-free spectral probe that posted a 34,200-step validated lead died under pre-registered confirmation — it false-alarmed on fresh structural negatives and, per-run, was largely detecting the *intervention*, not the emergence (kill K9). Both deaths are in the ledger (Section 6) and both shaped the LM protocol.

5 Part II: language models

5.1 Public checkpoints: the precursor leads, and why public suites are not enough

We probed 5 Pythia suites [17] (70m/160m/410m, standard and deduped) across training checkpoints for three quantities: behavioral copy advantage on repeated random sequences, the maximum per-head prefix-matching (induction) score, and the maximum early-layer previous-token score — the known mechanistic precursor [2]. The phase change is crisp in every suite (Figure 3): at 1.07B tokens copy advantage is 0.022 and the precursor is already at 0.37 ($\sim$10× baseline) while induction sits at 0.019; one stage later both cliff (copy 9.7 nats, induction 0.93). Against a best-case loss threshold granted an oracle sweep, the precursor alarmed strictly earlier on 4/5 suites and never post-event (leads 1–8 stages); our pre-registered prediction here failed in both directions — the precursor led by *more* than predicted and loss was *stronger* than predicted (one stage of genuine timing) — and is reported as registered. Two structural facts matter more than the scores. First, all five suites, which share batch and context configuration, cliff at the same $\sim$2.1B tokens — consistent with config-determined timing [1] and fatal to using public suites for per-seed forecasting: there is one run per config. Second, an event definition we had frozen (50% of each run's maximum) was silently broken by a late-training outlier in one suite — the same operationalization failure our grokking rungs had already punished — and was replaced, in print, by absolute criteria everywhere.

5.2 The seed fleet: per-seed forecasts, certified against manufactured negatives

Public suites cannot answer the forecasting question; fleets can. We trained 45 two-layer transformers on a fixed synthetic language (bigram-structured, with variable-offset within-context repetition so that positional shortcuts cannot substitute for induction — a shortcut our first design permitted, caught in smoke testing): 30 positives at one configuration (seeds vary; config fixed), plus 15 **manufactured negatives** — 10 *data-ablated* (identical language, zero repetition: induction has no training signal) and 5 *architectural* (one layer: induction is structurally impossible [2]). Events use an absolute criterion (copy advantage ≥ 2 nats, two consecutive evals).

Results, scored once against the pre-registered spec. Per-seed event times span 5,650–7,400 steps (spread 1.31×; the pre-registered bar of 1.3× was cleared by 0.01 — variance at fixed config is real but modest, and we do not dramatize it). The precursor's crossing time forecasts each seed's event: $\rho = 0.977$ (bootstrap CI [0.911, 0.995], $n = 30$), median lead 975 steps (825–1,125) — $\sim$15% of the run issued as advance notice, seed by seed, where the config equation predicts one constant. The best-case loss threshold ($\theta = 2.077$, granted an oracle sweep) *ties* the ranking ($\rho = 0.977$) — and forecasts nothing: its median lead is 50 steps (minimum -25), i.e., it detects the cliff in progress. Our pre-registered comparison metric (rank correlation alone) was therefore *wrong as designed*; the two-dimensional truth — equal correlation, 20× the lead — is reported alongside the registered failure, and (correlation, lead) pairs are now the protocol's required form. A second dissociation: the graded *behavioral* ramp that precedes the cliff carries no timing information ($\rho = -0.03$) — circuits time the transition; early behavior does not.

The negatives certified the false-alarm side. On data-ablated runs the precursor never formed (max layer-0 previous-token score 0.036–0.061 vs. threshold 0.10): bare precursor FA 0/10, conjunction FA 0/10, with pre-event alarms on 30/30 positives. Here our pre-registered mechanistic prediction was wrong and is reported: we expected the precursor to form anyway (predicting false alarms that would require a conjunctive gate); in fact a pure bigram language is solvable through the embedding pathway alone, so previous-token attention only pays where repetition exists. The manufactured negatives thus *certified* the precursor rather than trapping it — certification is the other thing negative sets are for — and constructing a language where the trap is real (where previous-token context pays independently of repetition) is the named follow-up. Architectural negatives: zero events, induction score ≤ 0.014 throughout, zero alarms.

5.3 Calibrated intervals

Single thresholds are fragile (a P5 lesson bought with a dead result); forecasts should be intervals with measured coverage. Split-conformal calibration (seeds 1–15) around the anchored point forecast $t_{\text{pv}} + 975$ yields intervals of median width 300 steps that covered 15/15 held-out seeds (nominal 90%) — issued 975 steps before the event. The identical machinery anchored at the loss crossing covers too — with 50 steps of lead. Anchoring, not calibration, is where warning time comes from.

5.4 The blind gate

Everything above is one configuration. The ship-gate: freeze the entire rule verbatim — alarm at layer-0 previous-token score ≥ 0.10; forecast the event in $[t_{\text{alarm}} + 825,\ t_{\text{alarm}} + 1125]$ — commit it, then train 10 fresh seeds and 3 fresh data-ablated negatives on a configuration the rule never saw (model width $256 \to 320$; repetition density $0.75 \to 0.6$), and score once. Pre-registered expectations were deliberately humble: the P5 transfer kill (fitted calibration did not survive regime shift) made the frozen offset the component we flagged as most at risk. Outcome (Figure 1): 10/10 events; 10/10 pre-event alarms; **10/10 interval coverage** against a bar of 7/10; $\rho = 0.988$ within the new config; 0/3 false alarms; median lead 1,012 steps. The precursor-to-emergence gap barely moved under the shift ($975 \to 1{,}012$ steps) — our flagged risk did not materialize, and we do not fully understand why: whether the gap is set by optimization dynamics rather than configuration is, we believe, a sharp open question this benchmark makes askable.

6 The kill ledger

Positive results are only as credible as the regime that tried to kill them. Everything below fired as pre-registered and is reported with its mechanism:

kill / failure	where	mechanism
K1/K8: no forecaster at 5% FA	grokking (algorithmic)	trap negatives out-signal true positives
K5: calibration non-transfer	grokking (cross-domain)	fitted thresholds are regime-local
K9: label-free probe dies	grokking (confirmation)	signal was the intervention, not emergence
P1 (both clauses)	Pythia	precursor earlier, loss stronger than predicted
50%-of-max event rules	twice	fragile to outliers; absolute criteria adopted
P2c metric as registered	fleet	rank correlation without lead rewards nowcasts
P2d interior mechanism	fleet	bigram languages need no attention; negatives certified
single-negative FA cap	grokking (R5)	degenerate fit; minimum-negative rule adopted

Three of these (K1/K8, K5, K9) are full kill criteria; the rest are registered predictions that failed informatively. The protocol that survived this ledger is the one the language-model results were scored under.

7 Discussion: what this does and does not establish

Established. For a named capability with a known mechanistic precursor, in small transformers: emergence timing is forecastable per run, $\sim$15% of training in advance, with calibrated intervals (15/15 and 10/10 coverage), certified false-alarm rates (0 across 18+3 manufactured negatives), a baseline accounting showing loss curves carry ranking but no lead, and blind transfer across a config shift. The evaluation discipline — manufactured negatives, (correlation, lead) pairs, FA caps with minimum negative counts, absolute event criteria, prereg-gated freezes — is domain-general and, we argue, the minimum standard any early-warning claim should meet.

Not established. Frontier applicability. The gap has named parts: (1) *precursor knowledge* — induction heads have a known antecedent; most capabilities of concern do not, and finding precursors is mechanistic-interpretability work our benchmark can score but not replace; (2) *negatives at scale* — frontier labs have n=1 runs and no capability-blocked counterfactuals; our manufactured-negative recipe (ablate the training signal the capability needs; block the architecture) is implementable in principle at any scale, and we regard a library of blocked runs as the single most actionable transfer of this work; (3) *recipe shift* — we passed one blind config shift on two axes; the P5 tradeoff says shift-robustness must be re-certified per monitor, and mechanism-factored designs are the only pattern we found that survives shift by construction; (4) *the gap-origin question* — the config-invariance of the precursor-to-cliff gap is unexplained; if it is an optimization-dynamics constant, calibration may transfer far more broadly than P5's kill suggests, and if not, per-config calibration fleets are the cost of coverage. Emergence forecasting in the wild also faces capabilities that *regress* (pythia-70m's late induction loss) and metric-induced discontinuities [3]; our absolute-criterion and interval machinery handles both mechanically, but the monitoring literature has not.

Practical recipe (what a lab could run tomorrow, at their scale, for a named capability): identify the precursor circuit; log its strength densely alongside a behavioral probe; manufacture blocked negatives by ablating the capability's training signal; calibrate an anchored conformal interval on a seed fleet; freeze; monitor; and re-certify after any recipe change — expecting, per our tradeoff, that the FA-control features are the first casualties of shift.

8 Limitations

One capability (induction), small models (2-layer fleets; $\leq$410m public suites), synthetic fleet language, one blind config shift (two axes bundled), n=30/10/18 per role. The fleet's positive-vs-negative separation was clean partly because a bigram language makes the precursor repetition-specific; harder languages may reintroduce the trap our grokking rungs documented. R4's calibration seeds were not blind (R5 is). The per-seed spread cleared its pre-registered bar by 0.01. Pythia event times are bounded by public checkpoint granularity. All pre-registration is repository-native (commit-stamped, publicly pushed), not third-party.

9 Reproducibility

Every number in this paper is a macro generated from scored artifacts (`paper4/gen_numbers.py`, byte-verified by `paper4/verify_regen.py`). The prereg chain is public: plan and kills before code (`2cf62e3`), forecaster freezes before test blocks, fleet spec before launch (`6b05770`), the blind gate frozen verbatim before its runs existed (`377511b`), verdicts after (`f25aa45`, `dd4d5f5`, `2ca7da4`). Corpora: 466 grokking runs, 5 public-suite probe trajectories, 45+13 fleet runs with dense circuit probes; training and scoring harnesses: `src/train_lm.py`, `src/probe_pythia.py`, `analysis/score_r*_p6.py`. Hardware: one RTX 3080.

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